

General Principles

1. A named entity contains, by definition, at least one capitalized word. Edge cases that could instead be coreferences should be treated with caution.
2. Entities are tagged only at surface level, without attempting an interpretation based on context. Metonymic or metaphoric readings (people-for-place, place-for-people) are not considered. Examples: 'I am going to Julia' (i.e. to Julia's house) Julia should be annotated as a **person**, not a location; 'Athens voted for the war': Athens is annotated as a **place**, even though it is used as a collective.
3. Boundaries: Anything that can be reduced to a single-word entity should be annotated as such. See the [specification on how to establish the boundaries of names](#) when they appear with multiple lowercase words.
4. Wherever possible, try to avoid annotating articles.

First-level tags

Person

Any identifiable single individual, including deities and anthropomorphic mythological figures. If the name appears with an epithet of ethnic, they should be annotated together. See further the [specification on how to annotate names and epithets](#).

Examples: 'Ajax', 'Ajax Telamonius', 'Thucydides the Athenian', etc.

Available second-level tags: **.epithet**, **.ethnic**, **.mythological**, **.astronomy**, **.author**, **.ancestry**

Location

A word or multiple words used to identify a politically, culturally, and/or geographically defined location, including geo-political, physical, fictional spaces and structures like temples, buildings, houses, and monuments.

When the name appears together with various lowercase words, these should be included when they disambiguate the entity type. See below for the [specification on how to establish the boundaries of names](#).

Examples: 'Athens', 'city of the Athenians', 'Ionian sea', 'temple of Apollo', 'Tartaros', etc.

Available second-level tags: **.epithet**, **.ethnic**, **.mythological**, **.astronomy**

Collective

A named group of people or other entities (creatures, animals, etc.) with shared identifiable characteristics on social, political, national, family, mythical, or ethnic basis.

Examples: 'Athenians', 'Cyclopes', 'Eumolpidae', 'Tarquinii', 'Senate', etc.

Available second-level tags: **.epithet**, **.ethnic**, **.animal**, **.mythological**, **.astronomy**, **.organization**, **.ancestry**.

Creature

Mythical or real precisely identifiable non-human, non-anthropomorphic creatures.

Examples: 'Bucephalus', 'Chimaera'.

Available second-level tags: **.animal**, **.mythological**, **.astronomy**

Event

Significant named events identified by a string with a precise boundary.

Examples: 'Peloponnesian war', 'Flood of Deucalion', 'Battle of Naupactus'.

Time

Any absolute date or time expression. Examples: '23rd Olympiad', '1985', 'Consulate of Cicero'.

Language

Languages clearly identified as such. Examples: 'Latin', 'English', 'Greek'.

Object

Artifacts or groups of artifacts identified with a name, such as ships, weapons, statues, columns, dedications, etc. Examples: 'Argos' (the ship of the Argonauts, not the person, the city, or Odysseus' dog).

Work

Titles of literary or non-literary works, in any form. Examples: 'Odyssey', 'History of the Peloponnesian War', 'Oedipus Rex'.

Miscellaneous

Entities that do not (yet) have a specific first-level tag among those provided.

Second-level tags

Second-level tags provide further specifications for particular types of Named Entities. They should be nested appropriately and according to the combinations described for each tag.

.epithet (person.epithet, collective.epithet, location.epithet)

A capitalized epithet used to refer unambiguously to one individual, location, or collective, including nicknames, titles, and other appellatives.

Example: 'Telamonius'.

Note: This tag should only be used when the epithet appears in isolation. If the epithet appears together with a personal name, e.g. ‘Ajax Telamonius’, they should be annotated together and labeled with the appropriate first-level tag (see specification about [personal names + epithets](#) below).

.ethnic (person.ethnic, collective.ethnic, location.ethnic)

Ethnonyms and demonyms used to refer unambiguously to one individual or collective by means of their association with a geographically identifiable territory.
In project-specific settings, these instances may also be associated to place information and coordinates for further study.

Examples: ‘(the) Athenian’: **person.ethnic**; ‘Athenians’: **collective.ethnic**.

Note: This tag should only be used when the ethnic appears in isolation. If the ethnic appears together with a personal name, e.g. ‘Thucydides the Athenian’, they should be annotated together and labeled with the appropriate first-level tag (see specification about [personal names + epithets](#) below).

.animal (collective.animal, creature.animal)

A type of creature or collective of creatures clearly identifiable with an animal or animal species.
Example: ‘cattle of Helios’, ‘Bucephalos’.

.mythological (creature.mythological, person.mythological, location.mythological, collective.mythological)

Any creature or group that belongs to the larger realm of myth and religion, including gods and deities. Examples: “Tartaros” (location.mythological), “Hercules” (person.mythological)

.astronomy (creature.astronomy, collective.astronomy, person.astronomy, location.astronomy)

Named stars, groups of stars, constellations, and planets.
Example: ‘Pleiades’

.organization (collective.organization)

Collectives identified by precise organizational structures, such as priesthoods, legions, religious or political groups, and so on.
Examples: ‘Eumolpidae’, ‘Senate’.

.author (person.author)

A person clearly mentioned in relation to works they have authored.

.ancestry (person.ancestry, collective.ancestry)

A name that refers unambiguously to one individual or group of individuals by using a family name, patronymic, matronymic, or other indication of ancestry. Examples: ‘Atreid’, ‘Tarquinius’.

Note: This tag should only be used when the ancestry appears in isolation. If the ancestry appears together with a name, e.g. ‘Achilles Pelides’, they should be annotated together and labeled with the appropriate first-level tag (see specification about [personal names + epithets](#) below).

Special cases and exceptions

Nested and multi-word entities

In the context of annotating a text, **multi-word entities are considered as single annotations** most of the time and tagged with the appropriate first-level tag. For example, ‘Thucydides the Athenian’ is tagged as a **person**.

Exceptions: a nested approach is acceptable when it is supported by a specific research goal. For example, you may want to investigate the provenance of all authors mentioned in a corpus using NE annotations. In this case, we recommend a max. 2-level nested annotation following Romanello-Najem-Meyer (2022):

```
<pers.author>Thucydides of <place>Athens</place></pers.author>  
<pers.author>Thucydides <pers.ethnic>the Athenian</pers.ethnic></pers.author>
```

Personal names + epithets, nicknames, appellatives

Whenever a personal name appears in a multi-word string that includes an epithet of some kind, **only cases where the epithet is capitalized are considered**. Where the epithet appears lowercase as an attribute to the name, only the personal name is annotated.

Exceptions: there are very few edge cases where an epithet appears lowercase in one edition but not in another to describe and disambiguate a personal name unequivocally. In these cases, it is necessary to double-check with dictionaries and authority lists, and the epithet may be annotated. For example, in the string ‘Apollon aigletes’, ‘aigletes’ (αἰγλήτης) is an epithet specific to Apollo often found capitalized.

Boundaries: When a Named Entity appears with lowercase words

By definition, a Named Entity is something that can be named. In cases where a capitalized name occurs with lowercase words, it can be tricky to establish the boundaries of what exactly constitutes a Named Entity.

In order to establish the boundaries of what constitutes a Named Entity in a text, try answering the following questions:

- 1) **What entity is actually being talked about?** If the lowercase word is necessary to appropriately label the entity the text refers to, then it should be included. Example: in the string “temple of Zeus”, the entity is not the **person** Zeus, but the **place** “temple of

Zeus". On the contrary, in "river Nile", only "Nile" is to be annotated because the word "river" is not necessary to identify it.

- 2) **Can the annotated string be associated with a precise named entity?** Example: the "sister of Augustus" is Octavia the Younger, so it can be annotated as a string and tagged **person**. This applies also to cases where the entity has an attested name, but its identity may not be known precisely due to lack of information.

What can be classified as multi-word entity:

1. "city of the Athenians" is a **place**. "Athenians" alone is an **ethnic**, but the entity the text talks about is a **place**, which can be associated with the name "Athens". Therefore, the entire string "city of the Athenians" is tagged as **place**.
2. "Black Sea" is a full string referring to a **place**. "Black" alone does not mean anything, and the entity the text talks about can be associated with a specific record in gazetteers.
3. "upper Egypt" is a **place**, as it refers to a precisely identifiable area that has a specific name (often all capitalized: Upper Egypt, vs. Lower Egypt).

What cannot be classified as a multi-word entity:

1. "Mediterranean sea": the word "sea" in this string does not further identify the entity "Mediterranean", nor does it change its label. Therefore, only "Mediterranean" should be tagged as **place**.
2. "Iberian coast": the Iberian coast is not a clearly identifiable place with its own name, and it cannot be associated with a precise identity record. Therefore, only "Iberian" should be tagged as **place** and associated with "Iberia".
3. "Region of Carthage": this is not a clearly identifiable place that can be named. Therefore, only "Carthage" should be annotated.

Exceptional non-capitalized named entities

There are a few edge cases of names that appear in lowercase, which however unequivocally refer to specific individuals, places, or groups. In these cases, these names should be annotated and tagged appropriately.

Examples of this kind are: 'symplegades' (i.e. the Symplegades), 'erynes' (the Furies), 'the hundred-handed' (the Hundred-Handed).

Unresolved issues and challenges

Non-consecutive multi-token entities

Strings that contain one named entity but are split across the text.

References

Authority lists useful for consultation

Theoi: <https://www.theoi.com/>

WikiData: https://www.wikidata.org/wiki/Wikidata:Main_Page

Trismegistos: <https://www.trismegistos.org/index.php>

Pleiades: <https://pleiades.stoa.org/>

Manto Project: <https://manto.unh.edu/>

ToposText: <https://topostext.org/index.php>

World Historical Gazetteer: <https://whgazetteer.org/search/#>

Platner, *Topographical dictionary of ancient Rome* [on Perseus](#)

Smith, *Dictionary of Greek and Roman Geography* (1854) [on Perseus](#)

Smith, *Dictionary of Greek and Roman biography and mythology* [on Perseus](#)

Smith, *Dictionary of Greek and Roman antiquities* (1890) [on Perseus](#)

Lexicon of Greek Personal Names: <https://www.lgpn.ox.ac.uk/>

Romans 1by1: <http://romans1by1.com/rpeople/people>

Bibliography

Romanello, M. and Najem-Meyer, S. (2022). Guidelines for the Annotation of Named Entities in the Domain of Classics (18.03.2022). Zenodo. <https://doi.org/10.5281/zenodo.6368101>

Hawes, G. (2023). 'The Manual'. Data Collection principles and practices. Manto Project.

Shipley, G. (2021). Sun, sea, and sky: on translating directions (and other terms) in the Greek Geographers. <https://hdl.handle.net/2381/13286000.v1>.

Shipley, G. (2024). "Introduction." In *Geographers of the Ancient World*. Vol. 1. Cambridge University Press.